

“ — — ”

100 8 92

$R^2=0.6351$ Adjusted $R^2=0.5986$ RMSE=68.1178

g MAE=34.8798 g 33.878° 30.612 mm 0.559 414.502

g

1

+ ” “ +

2

2.1

v d θ m

$$\hat{m} = \beta_0 + \beta_1 v + \beta_2 d + \beta_3 \theta + \beta_{11} v^2 + \beta_{22} d^2 + \beta_{33} \theta^2 + \beta_{12} v d + \beta_{13} v \theta + \beta_{23} d \theta.$$

$$\hat{m} = \mathbf{x}^T \mathbf{B} \mathbf{x} + \mathbf{b}^T \mathbf{x} + \beta_0, \quad \mathbf{x} = [v, d, \theta]^T.$$

2.2

$$\max_{v,d,\theta} \hat{m}(v,d,\theta)$$

$$v \in [0.2, 1.0], \quad d \in [20, 40], \quad \theta \in [20, 60].$$

2.3

m_i $\hat{m}_i$ n p

$$R^2 = 1 - \frac{\sum_{i=1}^n (m_i - \hat{m}_i)^2}{\sum_{i=1}^n (m_i - \bar{m})^2},$$

$$R^2_{adj} = 1 - (1 - R^2) \frac{n - 1}{n - p - 1},$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (m_i - \hat{m}_i)^2}, \quad MAE = \frac{1}{n} \sum_{i=1}^n |m_i - \hat{m}_i|.$$

2.4

<i>v</i>	
<i>d</i>	mm
<i>θ</i>	deg
<i>m</i>	g
<i>β</i>	-

“ ”

3

3.1

” “ — —

3.2

“ + ” 8 92 100

3.3

3.4

- 0.2–1.0
- 20–40 mm
- 20–60 deg

4

- 1
- 2 8
- 3 -

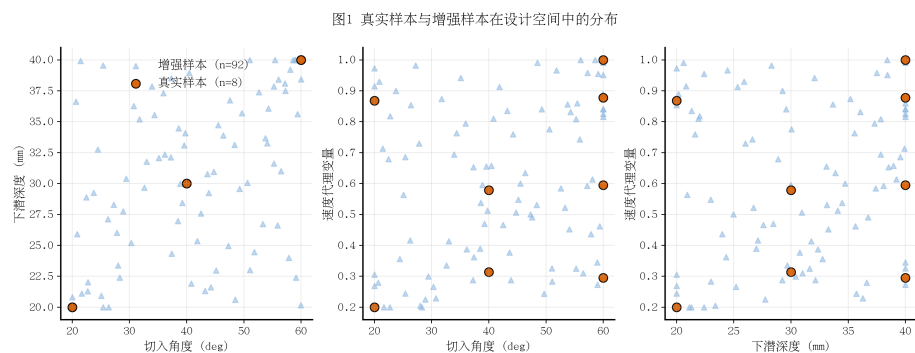


Figure 1: 1

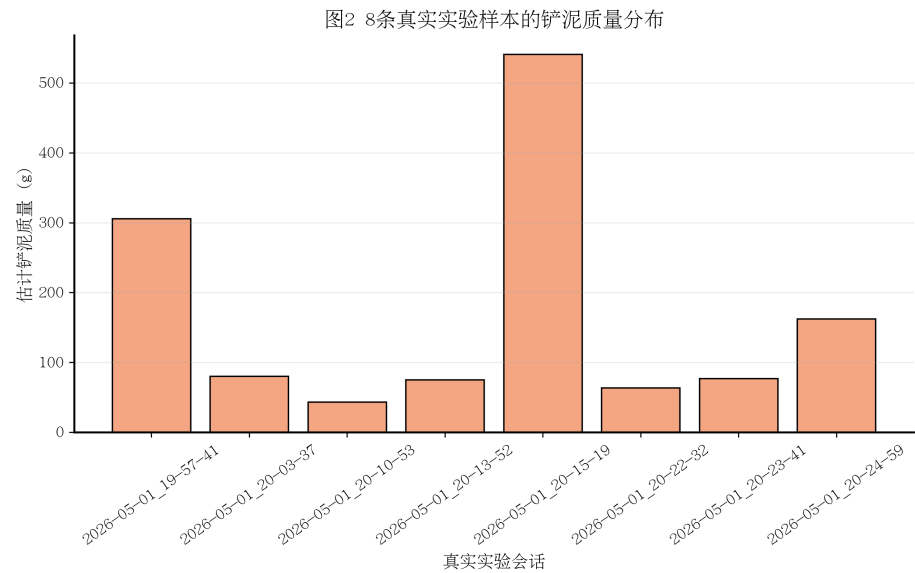


Figure 2: 2 8

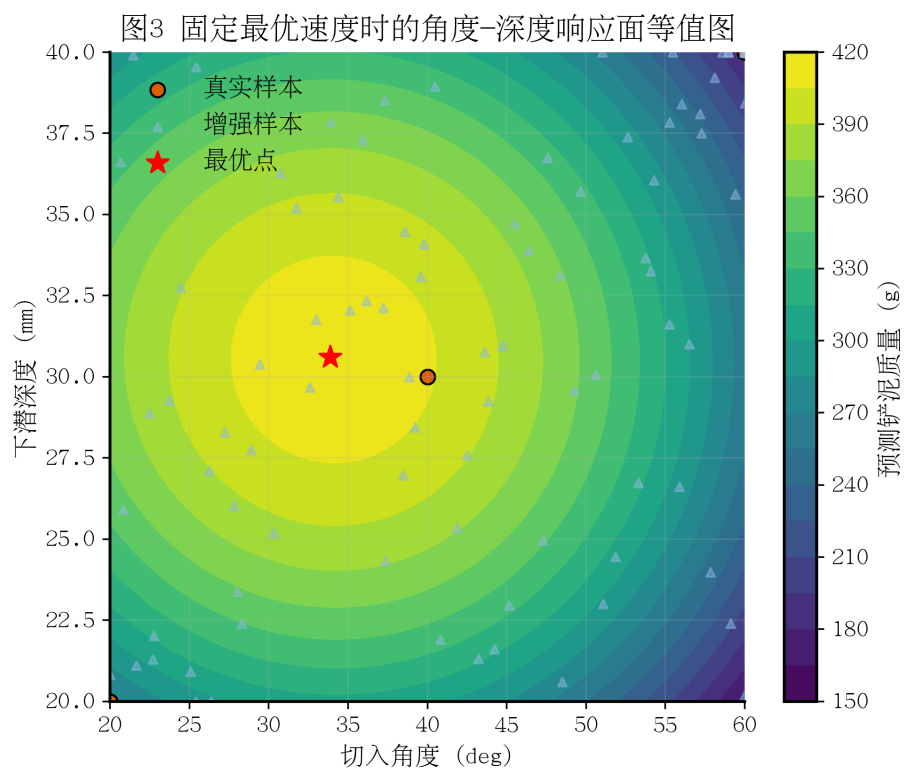


Figure 3: 3 -

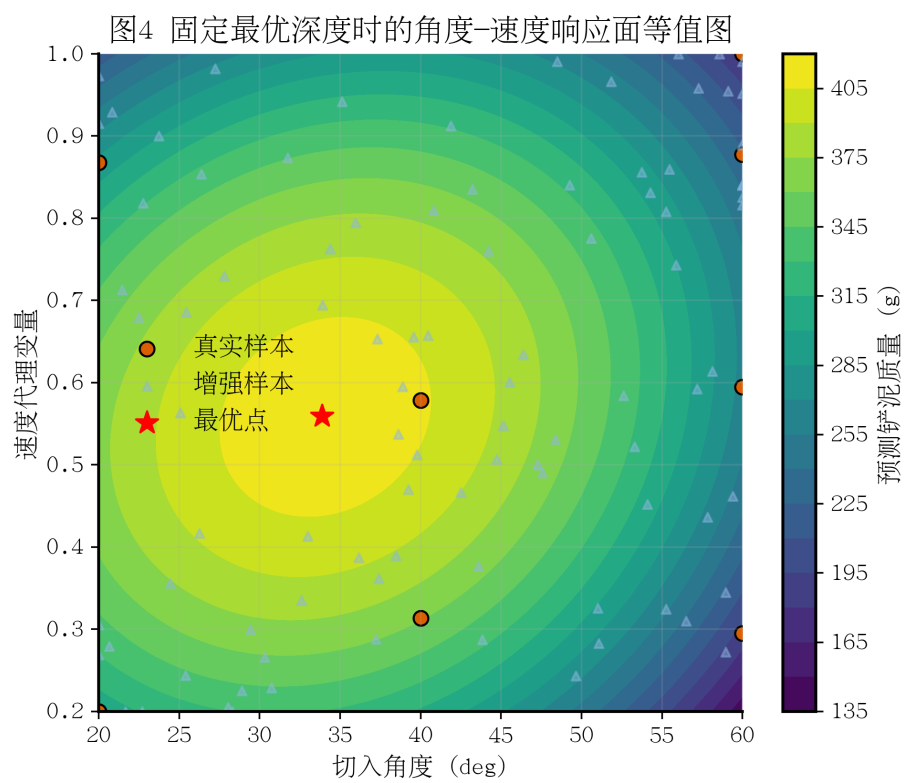


Figure 4: 4 -

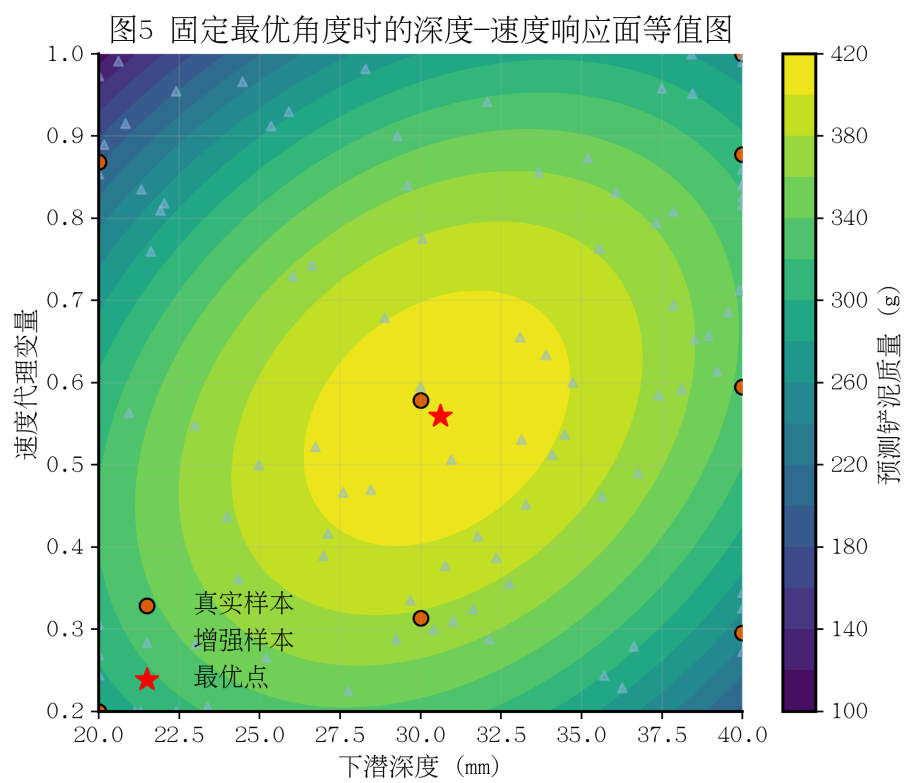


Figure 5: 5 -

4 -
5 -

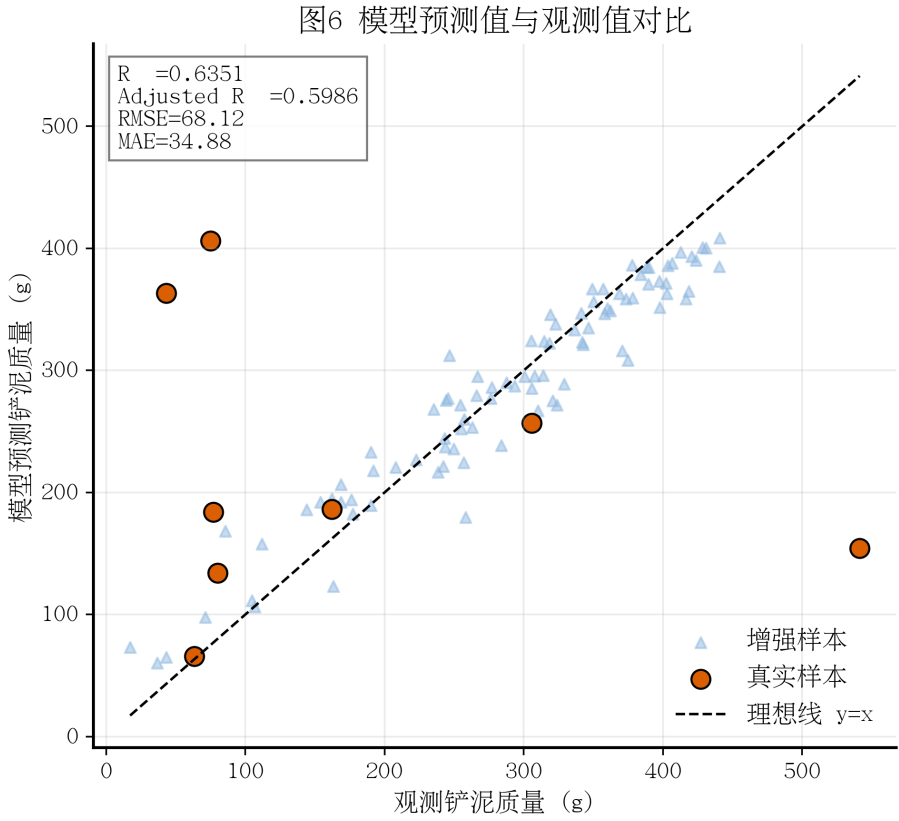


Figure 6: 6

6
7 -
8
9

4.1

$R^2=0.6351$ Adjusted $R^2=0.5986$ RMSE=68.1178 g MAE=34.8798 g

“ ”

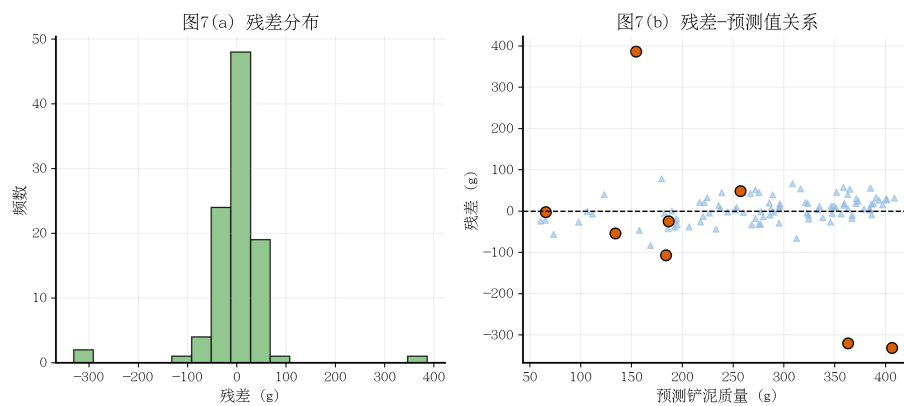


Figure 7: 7 -

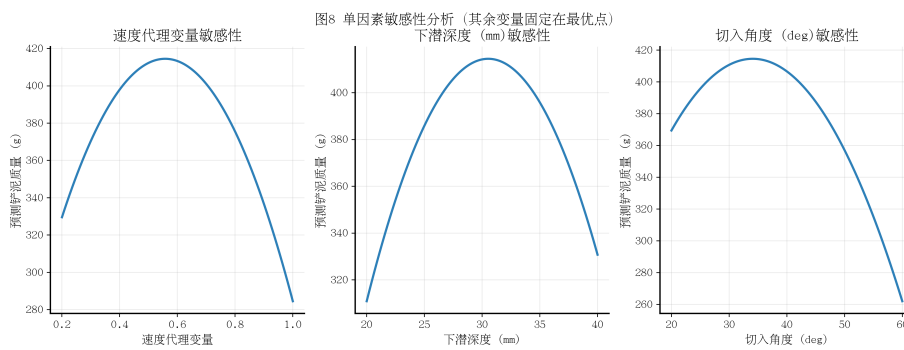


Figure 8: 8

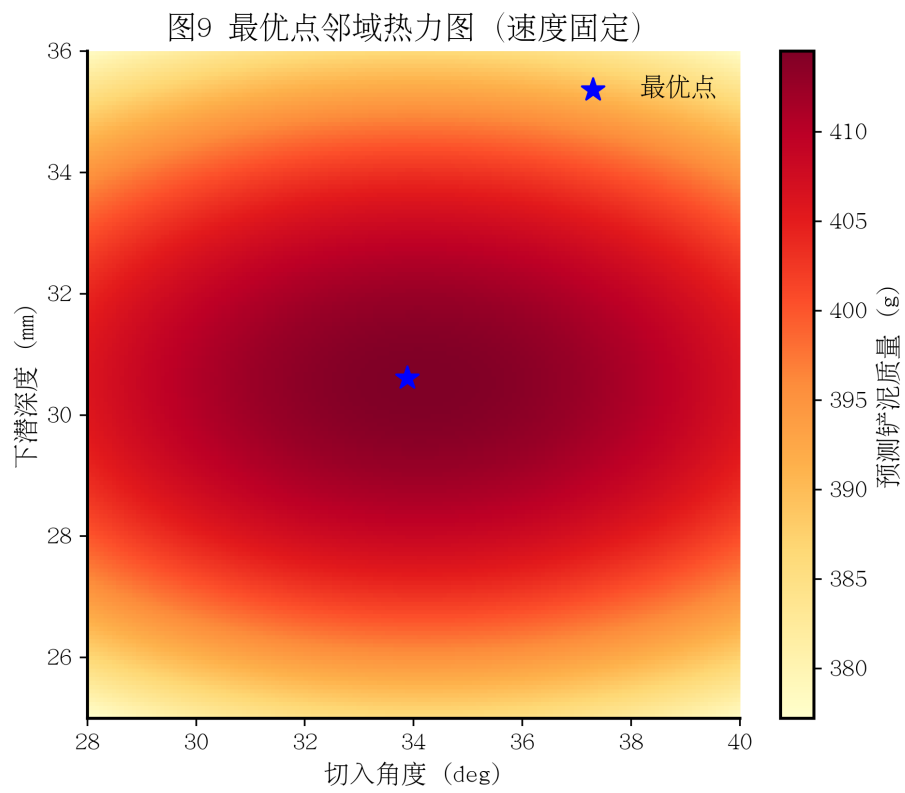


Figure 9: 9

4.2 1–5

1 2
3 5

4.3 6–9

6
7
8 “ ” 9

4.4

- 33.878 deg
- 30.612 mm
- 0.559
- 414.502 g

5

- 8
-
-
- “ ”

6

“ — — — ”

A

- 100 8 92
- R^2 0.6351

- Adjusted R^2 0.5986
- RMSE 68.1178 g
- MAE 34.8798 g
- 33.878 deg
- 30.612 mm
- 0.559
- 414.502 g

B